

7	(a) (i) progressive: energy is moved / transferred / propagated from one place to another (without the bulk movement of the medium)	B1
	transverse: (particles) oscillate / vibrate at right angles to the direction of travel of the energy / wavefront	B1 [2]
	(ii) number of oscillations per unit time / number of wavefronts passing a point per unit time	B1 [1]
	(b) (i) P and T	B1 [1]
	(ii) P and S <u>or</u> Q and T	B1 [1]
	(c) $\lambda = 1.2 \times 10^{-2} \text{ (m)}$	C1
	$v = f\lambda$ $= 15 \times 1.2 \times 10^{-2}$ $= 0.18 \text{ ms}^{-1}$	C1 A1 [3]
	(d) ratio = $(1.4)^2 / (2.1)^2$ = 0.44	C1 A1 [2]